JOYCE UNIVERSITY · BIO 280 PATHOPHYSIOLOGY

Exam 1 Master Study Guide

Modules 1 & 2 — Fluids, Electrolytes, ABGs, Pathogens & Immune Disorders

ⓘ What's on Exam 1

50 questions (multiple choice, select-all-that-apply). Proctored with Respondus Lockdown Browser. Closed book, no notes, no internet. Whiteboard allowed (show blank at start & end). **Available Week 3, Tuesday 12 AM — Thursday 11:59 PM.**

Unit exams are **50% of your overall grade** — treat this guide like a road map and practice with the hidden answers.

ⓘ How to use this guide

- **Click any topic card** to jump to it. **Click the title bar** to expand/close.
- **Answers are hidden** — try first, then click "Show answer." Active recall is what builds memory.
- **Each topic has:** content, mnemonic, common confusion, free-response Qs, multiple-choice Qs, clinical scenario, and a self-assessment checklist.
- **To save as PDF:** in your browser, press *Ctrl+P* (Cmd+P on Mac) → choose "Save as PDF." All collapsed sections will expand automatically for printing.

□14-Day Study Schedule — Built for Exam Week ▼

Day	Focus
Day 1–2	Topics 1–2 (fluid compartments, hormones). Memorize ICF vs ECF.
Day 3–4	Topic 3 (electrolytes). Memorize lab values for Na, K, Ca, P, Mg.
Day 5–6	Topics 4–5 (acid-base, buffer systems, acidosis/alkalosis).
Day 7	Topic 6 (ABG interpretation). Practice ALL 4 worked examples.
Day 8–9	Topics 7–9 (pathogens, infection stages, UTIs).
Day 10–11	Topics 10–11 (immunity, immunoglobulins, hypersensitivity, transplant).
Day 12	Topic 12 (HIV/AIDS, immunodeficiency, autoimmunity, leukemia/lymphoma).
Day 13	Re-do every "wrong" answer. Run the cheat sheet.
Day 14	Light review only. Sleep. Eat. Hydrate. **Take the exam.**

□Topic Navigator

TOPIC 1 · WEEK 1
Fluid Compartments & Pressures

TOPIC 2 · WEEK 1
Fluid Regulation Hormones

TOPIC 3 · WEEK 1
Electrolyte Imbalances

TOPIC 4 · WEEK 1
Acid-Base & Buffers

TOPIC 5 · WEEK 1
Acidosis & Alkalosis Causes

TOPIC 6 · WEEK 1
ABG Interpretation

TOPIC 7 · WEEK 2
Pathogens & Resistance

TOPIC 8 · WEEK 2
Infection Stages & TB

TOPIC 9 · WEEK 2
UTIs: Cystitis vs Pyelo

TOPIC 10 · WEEK 2

TOPIC 11 · WEEK 2

TOPIC 12 · WEEK 2

Immunity & Immunoglobulins | Hypersensitivity & Transplant | HIV, Immunodeficiency, Cancers

□Exam Cheat Sheet | □Lab Values | □Glossary

□Normal Lab Values You MUST Know

Value	Normal Range	Memory Hint
Sodium (Na^+)	135–145 mEq/L	Largest range; fluid balance
Potassium (K^+)	3.5–5.0 mEq/L	Cardiac; tight range
Calcium (Ca)	9.0–10.5 mg/dL	Muscle, nerves, clotting
Phosphate (P)	2.5–4.5 mg/dL	OPPOSITE of calcium
Magnesium (Mg)	1.3–2.1 mEq/L	Muscle relaxer; opposite of Ca
pH	7.35–7.45	<7.35 = acidosis; >7.45 = alkalosis
$PaCO_2$	35–45 mmHg	Respiratory acid (high = acidic)
HCO_3^-	22–26 mEq/L	Metabolic base (high = alkaline)
PaO_2	80–100 mmHg	Oxygenation status

TOPIC 1

□Fluid Compartments & Pressures

MUST KNOW ▼

Where Body Water Lives

Compartment	% of Body Fluid	Dominant Electrolyte
Intracellular (ICF)	~2/3 (~67%)	High **Potassium (K^+)**
Extracellular (ECF)	~1/3 (~33%)	High **Sodium (Na^+)**

The ECF further breaks down into:

- **Intravascular fluid** — inside blood vessels (plasma).
- **Interstitial fluid** — in tissue spaces around cells.
- **Transcellular fluid** — smaller spaces (CSF, joint fluid, peritoneal, etc.).

Hydrostatic vs. Oncotic Pressure

□HYDROSTATIC pressure

A *"pushing"* force generated by the blood inside vessels. **Pushes fluid OUT** of capillaries into tissues.

What raises it? High BP, heart failure, IV fluid overload, blocked venous return.

□ONCOTIC pressure

A *"pulling"* force created mainly by **albumin** (a plasma protein). **Pulls fluid INTO** the vessel.

What lowers it? Liver failure (less albumin made), malnutrition (no protein in), kidney disease (albumin lost in urine).

Causes of Peripheral Edema (fluid in tissues)

Edema = fluid escapes from vessels into tissue. Happens when:

- **Hydrostatic pressure goes UP** → pushed too hard out (heart failure, hypertension).
- **Oncotic pressure goes DOWN** → nothing pulling fluid back in (liver failure, malnutrition, kidney disease losing albumin).
- **Lymphatic blockage** → can't drain fluid back (lymphedema, post-mastectomy).
- **Increased capillary permeability** → vessel walls leaky (inflammation, sepsis, burns).

Memory aid: Hydrostatic = "Hose pushing out" · **Oncotic = "Owe me back" (albumin pulling)**. Edema = pushers win over pullers.

⚠ **Common confusion:** Sodium is the dominant electrolyte *outside* the cell (ECF). Potassium is dominant *inside* (ICF). Lots of students mix these up.

Free-Response Practice

1. What % of body fluid is intracellular vs extracellular, and which electrolyte dominates each?

   Show answer

   > ICF ~2/3 (high K^+). ECF ~1/3 (high Na^+).

2. Compare hydrostatic and oncotic pressure.

   Show answer

   > Hydrostatic = pushing force, pushes fluid OUT of vessels. Oncotic = pulling force from albumin, pulls fluid INTO vessels.

3. List 4 causes of peripheral edema.

   Show answer

   > Increased hydrostatic pressure (heart failure), decreased oncotic pressure (low albumin from liver failure / malnutrition / kidney disease), lymphatic obstruction, increased capillary permeability (inflammation/sepsis).

4. What's the difference between intravascular and interstitial fluid?

   Show answer

   > Intravascular = inside blood vessels (plasma). Interstitial = around tissue cells, outside vessels.

5. Why does a patient with cirrhosis develop edema and ascites?

   Show answer

   > The damaged liver produces less albumin → low oncotic pressure → fluid stays in tissues/abdomen instead of being pulled back into vessels.

Multiple Choice Practice

1. The dominant electrolyte inside cells is:
   A) Sodium · B) Potassium · C) Calcium · D) Chloride

   Show answer

   B. Potassium dominates ICF.

2. A patient with severe burns develops generalized edema. The most likely cause is:
   A) Lymphatic blockage · B) Increased capillary permeability · C) Low hydrostatic pressure · D) High oncotic pressure

   Show answer

   B. Burn injury makes capillaries leaky — fluid escapes into tissues.

3. Which protein is primarily responsible for oncotic pressure?
   A) Hemoglobin · B) Globulin · C) Albumin · D) Fibrinogen

   Show answer

   C. Albumin.

Clinical Scenario

A 68-year-old with congestive heart failure presents with 3+ pitting edema in both ankles and difficulty breathing. Lab shows normal albumin. Explain the mechanism of the edema and why the lung symptoms developed.

Show answer

Heart failure → blood backs up in the venous system → **increased hydrostatic pressure** in capillaries → fluid pushed out into tissues (peripheral edema). When backup is in the left heart, pulmonary capillary pressure rises and fluid leaks into alveoli → pulmonary edema (shortness of breath, crackles). Albumin is normal because the cause is hydrostatic, not oncotic.

☑ Self-Assessment Checklist

- I can identify ICF vs ECF percentages and dominant electrolytes.
- I can break ECF into intravascular, interstitial, and transcellular.
- I can describe hydrostatic and oncotic pressure and what each does.
- I can list at least 4 causes of peripheral edema.
- I can identify albumin's role in fluid balance.

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TOPIC 2

Fluid Regulation Hormones — RAAS, ADH, ANP/BNP

MUST KNOW

The RAAS System (Renin-Angiotensin-Aldosterone System)

What activates it: low blood pressure, low blood volume, or low sodium detected by the kidneys.

The cascade:

1. Kidneys release **renin** in response to low BP/volume.
2. Renin converts **angiotensinogen** (from liver) into **angiotensin I**.
3. ACE (angiotensin-converting enzyme) in the lungs converts angiotensin I to **angiotensin II**.
4. Angiotensin II does 2 main things:
    - **Vasoconstricts** blood vessels → raises BP.
    - Triggers the adrenal cortex to release **aldosterone**.

Aldosterone vs. ADH — Effects on Kidneys

Hormone	From	Kidney Effect	Net Effect
Aldosterone	Adrenal cortex	Reabsorbs **Na^+** (and water with it); excretes **K^+**	↑ BP, ↑ volume; ↓ K^+
ADH (Vasopressin)	Posterior pituitary	Reabsorbs **water** (no sodium)	↑ BP, ↑ volume; concentrates urine

ANP & BNP — The "Off-Switch"

ANP (atrial natriuretic peptide) is released by the *atria* of the heart; **BNP** (brain/B-type natriuretic peptide) is released by the *ventricles*. Both are released when the heart is stretched (high volume or high pressure).

Net effects: opposite of RAAS!

- Excrete **sodium** (natriuresis) and **water**.
- Dilate blood vessels.
- Decrease BP and volume.
- Inhibit renin and aldosterone.

Clinical use: elevated BNP is a marker of heart failure.

Angiotensin II — Main Actions

- Vasoconstriction → raises BP.
- Stimulates aldosterone release.
- Stimulates ADH release.
- Triggers thirst.

This is why **ACE inhibitors** ("–pril" drugs like lisinopril) are used to lower BP — they block this whole cascade.

Memory aids:

- **RAAS = "Raises pressure."** Low BP → activates RAAS → saves Na & water → BP goes up.
- **ANP/BNP = "Anti-pressure."** High pressure stretches the heart → ANP/BNP → dumps Na & water → BP goes down.
- **Aldosterone = "Aldo loves sodium."**
- **ADH = "Anti-Diuretic" → holds onto water.**

⚠ **Common confusion:** Aldosterone keeps Na+water; ADH keeps ONLY water (no Na). So in conditions with SIADH (too much ADH), serum Na drops (dilutional hyponatremia).

Free-Response Practice

1. What activates the RAAS system?

   Show answer

   Low BP, low blood volume, or low sodium detected by the kidneys.

2. List the 4 main actions of angiotensin II.

   Show answer

   Vasoconstriction; triggers aldosterone; triggers ADH; stimulates thirst.

3. Compare what aldosterone and ADH do at the kidney.

   Show answer

   Aldosterone reabsorbs sodium (and water with it) and excretes potassium. ADH reabsorbs water only (no sodium).

4. What are the net effects of ANP and BNP?

Show answer

> Excrete sodium and water, dilate vessels, decrease BP and volume, inhibit RAAS. Opposite of RAAS.

5. Why is BNP elevated in heart failure?

   Show answer

   > Failing ventricles are overstretched by backed-up fluid, which triggers BNP release. Higher BNP = more stretch = more severe failure.

Multiple Choice Practice

1. Aldosterone causes the kidneys to:
   A) Excrete sodium · B) Reabsorb sodium and water; excrete potassium · C) Reabsorb water only · D) Dilate vessels

   Show answer

   > **B.** Reabsorbs Na (and water) & excretes K.

2. An elevated BNP level most directly suggests:
   A) Liver failure · B) Dehydration · C) Heart failure · D) Diabetes

   Show answer

   > **C.** Heart failure (ventricular stretch).

3. Which hormone primarily controls water reabsorption WITHOUT affecting sodium?
   A) Aldosterone · B) ADH · C) Angiotensin II · D) Renin

   Show answer

   > **B.** ADH = water only.

Clinical Scenario

A trauma patient has lost a significant amount of blood. The body's compensatory response includes which RAAS-related steps, and what would you expect to see in vital signs and urine output before treatment?

Show answer

Low volume → kidneys release renin → angiotensin II → vasoconstriction (BP rises a bit; HR rises) and aldosterone + ADH release (kidneys hold onto Na and water). Expected findings: tachycardia, narrow pulse pressure, **decreased urine output** (oliguria), and concentrated urine. Fluid resuscitation reverses the cascade.

☑ Self-Assessment Checklist

- I can list what triggers RAAS and trace renin → angiotensin II.
- I can name the 4 main actions of angiotensin II.
- I can contrast aldosterone (Na + water + K excretion) with ADH (water only).
- I can describe the opposite role of ANP/BNP.
- I can explain why BNP is a heart failure marker.

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TOPIC 3

□Electrolyte Imbalances (Na, K, Ca, P, Mg) MUST KNOW ▼

Hint: Water follows sodium. Potassium is "the cardiac one." Phosphate and magnesium move *opposite* to calcium.

⚒ Sodium (Na^+): 135–145 mEq/L

Function: fluid balance (kidneys control it).

	Hyponatremia (<135)	Hypernatremia (>145)
Causes	SIADH, diuretics, vomiting/diarrhea, water overload, heart failure (dilutional)	Dehydration, diabetes insipidus, excessive Na intake, burns
S/Sx	Headache, confusion, lethargy, seizures, muscle cramps. Brain swells (cells take in water).	Thirst, dry mucous membranes, irritability, lethargy, seizures. Cells shrink.

❤ Potassium (K^+): 3.5–5.0 mEq/L — "the cardiac one"

Function: cardiac contractility & acid-base regulation.

	Hypokalemia (<3.5)	Hyperkalemia (>5.0)
Causes	Diuretics (esp. loop), vomiting/diarrhea, alkalosis, low intake, NG suction	Kidney failure, acidosis, K-sparing diuretics, tissue damage (burns/crush), Addison's
S/Sx	Muscle weakness, fatigue, leg cramps, constipation, **flat T waves & U waves on EKG**, dysrhythmias	Muscle weakness, paresthesias, **peaked T waves on EKG**, dysrhythmias, cardiac arrest

□Calcium (Ca): 9.0–10.5 mg/dL

Function: muscle contraction, neurotransmitter release, clotting, bone strength.

	Hypocalcemia (<9.0)	Hypercalcemia (>10.5)
Causes	Hypoparathyroidism, vitamin D deficiency, kidney disease, malabsorption, pancreatitis	Hyperparathyroidism, cancer (bone metastases), excessive vitamin D, immobility

S/Sx	Muscle twitching, tetany, **Trousseau's sign** (carpal spasm), **Chvostek's sign** (facial twitch), paresthesias, seizures	Muscle weakness, bone pain, kidney stones, constipation, lethargy. "Bones, stones, groans, psychic moans."

□Phosphate (P): 2.5–4.5 mg/dL — *opposite of calcium*

Function: membrane stability, bone & teeth, cell energy, DNA/RNA.

	Hypophosphatemia (<2.5)	Hyperphosphatemia (>4.5)
Causes	Alcoholism, burns, refeeding syndrome, malabsorption	Kidney failure (most common)
S/Sx	Muscle pain & weakness, irritability, constipation, confusion	Muscle cramps, bone pain, pruritus (itching); s/sx mirror hypocalcemia (Ca drops as P rises)

□Magnesium (Mg): 1.3–2.1 mEq/L — "the muscle relaxer"

Function: muscle relaxant, potassium regulation. Acts *opposite* of calcium.

	Hypomagnesemia (<1.3)	Hypermagnesemia (>2.1)
Causes	Alcoholism, malabsorption, diarrhea, diuretics	Kidney failure, antacids/laxatives with Mg, IV Mg in preeclampsia
S/Sx	Looks like hypocalcemia — tetany, twitching, Trousseau's/Chvostek's, dysrhythmias, seizures	Lethargy, hypotension, **depressed deep tendon reflexes**, respiratory depression, cardiac arrest

□Master pattern:

- **Calcium ↔ Phosphate**: one up, the other down.
- **Calcium ↔ Magnesium**: function-wise opposite, but their lab values often move together (both deficient in alcoholism, both excessive in kidney failure).
- **Hypocalcemia s/sx = Hypomagnesemia s/sx**: tetany, Trousseau's, Chvostek's.
- **EKG changes**: hyperkalemia = peaked T; hypokalemia = flat T & U waves.

⚠ **Common confusion:** "Hypo" and "Hyper" can cause the SAME symptom (e.g., muscle weakness in BOTH hyperkalemia and hypokalemia). Pin down the lab value and the specific signs (peaked vs flat T waves) to differentiate.

Free-Response Practice

1. State the normal lab range for Na, K, Ca, P, and Mg.

   Show answer

   Na: 135–145 mEq/L. K: 3.5–5.0 mEq/L. Ca: 9.0–10.5 mg/dL. P: 2.5–4.5 mg/dL. Mg: 1.3–2.1 mEq/L.

2. What EKG change classically appears in hyperkalemia vs hypokalemia?

   Show answer

   Hyperkalemia: peaked T waves. Hypokalemia: flattened T waves with prominent U waves.

3. Name Trousseau's and Chvostek's signs and what electrolyte imbalance they indicate.

   Show answer

   Trousseau's = carpal spasm after BP cuff is inflated. Chvostek's = facial twitch when cheek is tapped. Both indicate hypocalcemia (or hypomagnesemia).

4. Why does kidney failure tend to cause hyperphosphatemia AND hypocalcemia?

   Show answer

   Failing kidneys can't excrete phosphate → P builds up. Calcium and phosphate move in opposite directions, so Ca drops as P rises. Failed kidneys also activate less vitamin D, so Ca absorption falls further.

5. What are signs of magnesium toxicity (hypermagnesemia)?

   Show answer

   Depressed deep tendon reflexes, lethargy, hypotension, respiratory depression, cardiac arrest. (Why nurses check reflexes during IV Mg for preeclampsia.)

□Multiple Choice Practice (SATA practice too)

1. Which lab finding is most consistent with hyperkalemia?
   A) Flat T waves · B) Peaked T waves · C) Prolonged QT · D) U waves

   Show answer

   B. Peaked T waves.

2. Select ALL that apply — causes of hypokalemia:
   A) Loop diuretics · B) Vomiting · C) Kidney failure · D) Alkalosis · E) Acidosis

   Show answer

   A, B, D. Loop diuretics waste K, vomiting loses K, and alkalosis shifts K into cells (low serum K). Kidney failure and acidosis cause hyper-K, not hypo.

3. A patient with hypocalcemia is likely to show:
   A) Bone pain · B) Constipation · C) Trousseau's sign · D) Lethargy

   Show answer

   C. Trousseau's sign — carpal spasm with BP cuff inflation.

□Clinical Scenario

A patient with chronic kidney disease has the following labs: K^+ 6.1, Ca 7.8, P 6.5. The patient feels weak and reports tingling around the mouth. Which two abnormal labs are likely connected, and what does the EKG most likely show?

Show answer

Ca (7.8 = low) and P (6.5 = high) are connected: failing kidneys can't excrete P, and Ca/P move opposite. Tingling = hypocalcemia. K (6.1 = high) is from inability to excrete K. EKG would show **peaked T waves**. Treatment may include calcium gluconate (stabilizes the heart) and measures to lower K (insulin/D50, kayexalate, dialysis).

☑ Self-Assessment Checklist

- I can recite the normal ranges for Na, K, Ca, P, Mg from memory.
- I can list at least 3 causes and 3 s/sx for each high and low electrolyte.
- I can identify the EKG findings for hypo- and hyperkalemia.
- I can describe Trousseau's and Chvostek's signs.
- I can explain why kidney failure links high P, low Ca, and high K.

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TOPIC 4

⚖ Acid-Base Foundations & Buffer Systems

MUST KNOW

Acid-Base Basics

- Normal metabolism creates **CO_2** and **H^+** — both are acids.
- Body's goal: keep pH between **7.35–7.45** (very narrow range).
- **pH and H^+ are inversely proportional**: if H^+ is high, pH is low (acidic).
- **Acids:** CO_2, H^+. **Base:** HCO_3^-.

The 3 Buffer Systems

Buffer	Speed	How It Works
Serum (chemical)	Immediate (seconds)	Proteins bind H^+; K^+ exchanges with H^+ across cell membranes
Respiratory	Minutes	Lungs adjust ventilation. ↑ respiration blows off CO_2 → pH rises. ↓ respiration retains CO_2 → pH falls. Kussmaul = deep, rapid breaths (DKA)
Renal	2–3 days — *STRONGEST*	Kidneys excrete H^+ and reabsorb/produce HCO_3^-

Why K^+ Shifts With pH

In acidosis, H^+ floods into cells (to buffer) and K^+ gets pushed out into the blood → **acidosis = hyperkalemia**. The reverse in alkalosis → **alkalosis = hypokalemia**.

Memory aids:

- **CO_2 = ACID**. High CO_2 → acidic.
- **HCO_3^- = BASE**. High HCO_3^- → alkaline.
- "Acidosis = K up." "Alkalosis = K down."

⚠ **Common confusion:** Lower pH = MORE acid, not less. The smaller number means more H^+.

Free-Response Practice

1. State all 4 normal ABG values.

Show answer

> pH 7.35–7.45 · $PaCO_2$ 35–45 mmHg · HCO_3^- 22–26 mEq/L · PaO_2 80–100 mmHg.

2. Which is acid and which is base: CO_2 or HCO_3^-?

   Show answer

   > CO_2 = acid. HCO_3^- = base.

3. Rank the 3 buffer systems by speed.

   Show answer

   > Serum (seconds) → Respiratory (minutes) → Renal (days, strongest).

4. What is Kussmaul breathing?

   Show answer

   > Deep, rapid breathing that blows off CO_2 to compensate for metabolic acidosis (classic in DKA).

5. How does acidosis affect serum K^+?

   Show answer

   > Causes hyperkalemia — H^+ moves into cells, K^+ pushed out.

Multiple Choice Practice

1. Which buffer is fastest?
   A) Serum · B) Respiratory · C) Renal · D) Lymphatic

   Show answer

   > **A.** Serum (chemical) — works in seconds.

2. The strongest buffer system is:
   A) Serum · B) Respiratory · C) Renal · D) Cardiac

   Show answer

   > **C.** Renal — slowest but strongest.

3. pH 7.30 + $PaCO_2$ 50 indicates:
   A) Metabolic acidosis · B) Respiratory acidosis · C) Metabolic alkalosis · D) Respiratory alkalosis

Show answer

B. Acidic pH + high CO_2 = respiratory acidosis.

Clinical Scenario

A DKA patient is breathing fast and deep. Which buffer system is engaged and why?

Show answer

DKA causes metabolic acidosis (ketones). The **respiratory buffer** hyperventilates to blow off CO_2 (acid) and raise pH. Deep rapid breathing = Kussmaul respirations.

☑ Self-Assessment Checklist

- I know all 4 ABG normal values from memory.
- I can identify CO_2 as acid, HCO_3^- as base.
- I can rank the 3 buffer systems by speed and identify which is strongest.
- I can predict K^+ direction with acidosis vs alkalosis.

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TOPIC 5

□Acidosis & Alkalosis Causes

MUST KNOW ▼

Acidosis (pH < 7.35)

Type	Causes	Key Findings
Metabolic acidosis	DKA (ketones) · lactic acidosis · liver/kidney failure · severe diarrhea (loses HCO_3^-)	High anion gap · **hyperkalemia** · Kussmaul compensation
Respiratory acidosis	Hypoventilation: COPD, pneumonia, opioid overdose, sedation	**Hypercapnia** (↑CO_2) · **hyperkalemia**

Alkalosis (pH > 7.45)

Type	Causes	Key Findings
Metabolic alkalosis	Severe vomiting · NG suction (lose H^+) · excess calcium carbonate (antacids)	**Hypokalemia**
Respiratory alkalosis	Hyperventilation: panic, anxiety, fever, pain, mechanical over-ventilation	**Hypocapnia** (↓CO_2) · **hypokalemia**

Consequences of Acidosis on the Body

- Neuro: confusion, lethargy, coma.
- Cardiac: dysrhythmias, ↓contractility, vasodilation, hypotension.
- Respiratory: Kussmaul breathing (compensation).
- Bone: chronic acidosis pulls Ca out of bone.
- Electrolytes: hyperkalemia.
- Enzymes: denature at extreme pH.

□Quick patterns:

- **Vomiting/NG suction = ALKALOSIS** (lose gastric acid).
- **Diarrhea = ACIDOSIS** (lose intestinal HCO_3^-).
- **DKA = METABOLIC ACIDOSIS.**
- **COPD/opioid = RESPIRATORY ACIDOSIS.**
- **Panic attack = RESPIRATORY ALKALOSIS.**

⚠ **Common confusion:** Direction of GI loss matters. Upper GI (vomiting) loses acid = alkalosis. Lower GI (diarrhea) loses base = acidosis.

□Free-Response Practice

1. List 2 causes of metabolic acidosis and 2 of respiratory acidosis.

   Show answer

   Metabolic: DKA, severe diarrhea. Respiratory: COPD, opioid overdose.

2. List 2 causes of metabolic alkalosis and 2 of respiratory alkalosis.

   Show answer

   Metabolic: severe vomiting, NG suction. Respiratory: panic attack, fever-induced hyperventilation.

3. What K^+ pattern do you expect in each acid-base disorder?

   Show answer

   Acidosis → hyperkalemia. Alkalosis → hypokalemia.

4. List 3 consequences of acidosis on the body.

   Show answer

   Confusion/coma, dysrhythmias, hyperkalemia, Kussmaul compensation, bone Ca loss (chronic).

5. What does a high anion gap indicate?

   Show answer

   Metabolic acidosis with accumulated unmeasured acid — DKA, lactic acidosis, salicylates, renal failure.

□Multiple Choice Practice

1. Severe vomiting most likely causes:
   A) Metabolic acidosis · B) Metabolic alkalosis · C) Respiratory acidosis · D) Respiratory alkalosis

   Show answer

   B. Loss of HCl from stomach → metabolic alkalosis.

2. An opioid overdose patient is at greatest risk for:

   A) Metabolic acidosis · B) Metabolic alkalosis · C) Respiratory acidosis · D) Respiratory alkalosis

   Show answer

   C. Hypoventilation → CO_2 retention.

3. Severe diarrhea most likely causes:

   A) Metabolic acidosis · B) Metabolic alkalosis · C) Respiratory acidosis · D) Respiratory alkalosis

   Show answer

   A. Lose HCO_3^- from intestines → metabolic acidosis.

Clinical Scenario

A COPD patient comes in with worsening dyspnea. ABG: pH 7.30, $PaCO_2$ 60, HCO_3^- 30. Identify the disorder, the compensation effort, and expected K^+.

Show answer

Acidic pH; high CO_2 (matches); high HCO_3^- (kidney compensation). **Respiratory acidosis with partial metabolic compensation**. Expect **hyperkalemia**. Treat the underlying cause — improve ventilation, bronchodilators, careful O_2.

☑ Self-Assessment Checklist

- I can list 2 causes for each of the 4 acid-base disorders.
- I can predict K^+ direction in acidosis and alkalosis.
- I can list 3 consequences of acidosis.
- I can match a clinical condition to its likely acid-base disorder.

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TOPIC 6

□ABG Interpretation — Step-by-Step with Worked Cases

MUST KNOW ▼

Why we draw ABGs

To assess oxygenation (PaO_2), ventilation ($PaCO_2$), and acid-base balance — especially critical for ICU, vent management, and metabolic emergencies.

3-Step ABG Method

1. **Look at pH:** <7.35 = acidosis. >7.45 = alkalosis. 7.35–7.45 = normal (may be compensated).
2. **Decide respiratory or metabolic:**
   - Check **$PaCO_2$** (respiratory). >45 = acidic. <35 = alkalotic.
   - Check **HCO_3^-** (metabolic). <22 = acidic. >26 = alkalotic.
   - The value matching the pH direction is the CAUSE.
3. **Check for compensation:** if the OTHER value is also abnormal but moving OPPOSITE the pH, the body is compensating.

ROME Memory Trick

ROME	Meaning
Respiratory	**O**pposite — pH and CO_2 go OPPOSITE directions
Metabolic	**E**qual — pH and HCO_3^- go SAME direction

Compensation Levels

- **Uncompensated** — only the causative value is abnormal.
- **Partially compensated** — both values abnormal; pH still out of range.
- **Fully compensated** — both values abnormal; pH back in normal range.

Worked Examples (from your lecture slides)

Case 1: pH 7.29 · CO_2 47 · HCO_3^- 24

Step 1: pH 7.29 → acidic.

Step 2: CO_2 47 = high (acidic). HCO_3^- 24 = normal. CO_2 matches pH → **respiratory cause**.

Step 3: HCO_3^- is normal → no compensation yet.

Diagnosis: Uncompensated Respiratory Acidosis. Think: hypoventilation, opioid OD, acute COPD flare.

Case 2: pH 7.25 · CO_2 49 · HCO_3^- 30

Step 1: pH 7.25 → acidic.

Step 2: CO_2 49 = high (acidic). HCO_3^- 30 = high (alkaline). CO_2 matches pH → respiratory.

Step 3: HCO_3^- is high — kidneys trying to compensate. pH still <7.35.

Diagnosis: Respiratory Acidosis with Partial Metabolic Compensation. Classic chronic COPD picture.

Case 3: pH 7.31 · CO_2 30 · HCO_3^- 18

Step 1: pH 7.31 → acidic.

Step 2: CO_2 30 = low (alkaline). HCO_3^- 18 = low (acidic). HCO_3^- matches pH → **metabolic cause**.

Step 3: CO_2 is low — lungs hyperventilating to compensate. pH still acidic.

Diagnosis: Metabolic Acidosis with Partial Respiratory Compensation. Classic DKA — Kussmaul breathing.

Case 4: pH 7.48 · CO_2 41 · HCO_3^- 29

Step 1: pH 7.48 → alkaline.

Step 2: CO_2 41 = normal. HCO_3^- 29 = high (alkaline). HCO_3^- matches pH → metabolic.

Step 3: CO_2 normal → no respiratory compensation yet.

Diagnosis: Uncompensated Metabolic Alkalosis. Think: severe vomiting, NG suction, antacid overuse.

Tic-tac-toe shortcut: Make 3 columns — ACID / NORMAL / BASE. Drop each value (pH, CO_2, HCO_3^-) in the right column. Two in ACID = acidosis. The value alongside pH tells you the type. Remember: **CO_2 goes opposite to its column intuition** — high CO_2 is ACID, low CO_2 is BASE.

⚠ **Common confusion:** "Compensation" never overshoots back to normal in a single direction. If pH is normal AND both CO_2 and HCO_3^- are abnormal, it's *fully compensated* — the original cause is wherever pH is closer to (just barely) crossed the line.

Free-Response Practice

1. What does ROME stand for?

   Show answer

   > Respiratory Opposite, Metabolic Equal.

2. Interpret: pH 7.50, CO_2 28, HCO_3^- 24.

   Show answer

   > Alkalotic pH; low CO_2 (alkalotic) matches; normal HCO_3^-. **Uncompensated respiratory alkalosis** — think hyperventilation/panic.

3. Interpret: pH 7.32, CO_2 42, HCO_3^- 17.

   Show answer

   > Acidic pH; normal CO_2; low HCO_3^- (acidic) matches. **Uncompensated metabolic acidosis**.

4. Why is high CO_2 an acid?

   Show answer

   > $CO_2 + H_2O \rightarrow H_2CO_3 \rightarrow H^+ + HCO_3^-$. More CO_2 pushes reaction to produce more H^+.

5. What does a normal pH with abnormal CO_2 AND HCO_3^- mean?

   Show answer

   > Fully compensated state — body restored pH but both systems are still off-balance.

Multiple Choice Practice

1. pH 7.20, CO_2 50, HCO_3^- 22. This is:
   A) Uncompensated respiratory acidosis · B) Compensated metabolic acidosis · C) Metabolic alkalosis · D) Mixed disorder

   Show answer

   > **A.** Acidic pH; CO_2 high matches; HCO_3^- normal → uncompensated.

2. pH 7.42, CO_2 52, HCO_3^- 32. This is:
   A) Uncompensated respiratory acidosis · B) Fully compensated respiratory acidosis · C) Metabolic alkalosis · D) Normal

   Show answer

   > **B.** pH on low-normal side; both CO_2 and HCO_3^- abnormal → fully compensated respiratory acidosis. Classic chronic COPD.

3. A panic-attack patient most likely has:
   A) Respiratory acidosis · B) Respiratory alkalosis · C) Metabolic acidosis · D) Metabolic alkalosis

   Show answer

   > **B.** Hyperventilation → low CO_2 → respiratory alkalosis.

Clinical Scenario

A 22-year-old with Type 1 diabetes presents with confusion, Kussmaul respirations, and fruity breath. ABG: pH 7.18, CO_2 22, HCO_3^- 10. Interpret and identify the underlying problem.

Show answer

pH very acidic. CO_2 very low (alkalotic). HCO_3^- very low (acidic). HCO_3^- matches pH → metabolic acidosis. Low CO_2 = respiratory compensation (Kussmaul). pH still abnormal → **metabolic acidosis with partial respiratory compensation**. Underlying: **DKA**. Treatment: IV insulin, fluids, K replacement.

☑ Self-Assessment Checklist

- I can apply the 3-step ABG method consistently.
- I can use ROME to identify respiratory vs metabolic causes.
- I can distinguish uncompensated, partially compensated, and fully compensated.
- I can interpret all 4 worked examples without looking.
- I can match clinical conditions (DKA, COPD, vomiting, panic) to their ABG patterns.

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TOPIC 7

□Pathogens & Antibiotic Resistance

MUST KNOW ▼

Types of Pathogens

Pathogen	Key Features	Examples
Bacteria	Single-cell organisms; can replicate independently	Strep, Staph, TB, E. coli
Viruses	Depend on host cell metabolism; can't replicate alone	HIV, flu, COVID, herpes
Fungi	Spore-producing; superficial (skin) or deep (organs)	Candida (thrush, yeast), Tinea (ringworm)
Parasites	Helminths (worms), protozoa, insects; benefit at host's expense	Malaria, tapeworm, lice, scabies

Fungal Infections — Superficial vs Deep

Type	Where	Examples
Superficial	Mouth, skin, vagina	Candida (thrush, yeast infection), Tinea (ringworm)
Deep	Internal organs — sign of immune issue	Systemic candidiasis, cryptococcal meningitis

Many fungi are **normal flora**. Infections occur when they overgrow or invade (often after antibiotics wipe out competing flora).

Testing for Microbes

- **Gram stain** (bacteria):
  - **Gram-positive** — thick cell wall, stains **PURPLE**.
  - **Gram-negative** — thin cell wall, stains **PINK/RED**. (Harder to treat — outer membrane is a barrier.)
- **Culture & Sensitivity** — Culture identifies the organism; sensitivity shows which antibiotic will work.

Antibiotic Resistance — Contributing Factors

- Inappropriate antibiotic use (prescribed for viral infections like the cold/flu).
- Not finishing the full course of antibiotics — surviving bacteria multiply with resistance.
- Widespread use in agriculture & livestock.
- Patient non-adherence.
- Overuse of broad-spectrum antibiotics.
- Hospital-acquired exposure (MRSA, VRE, C. diff).

Gram-stain trick: **P**ositive = **P**urple. (Negative stains pink/red — the opposite of purple.) Gram-negatives tend to be harder to treat because their outer membrane keeps drugs out.

⚠ **Common confusion:** Antibiotics work ONLY on bacteria — not viruses. Giving antibiotics for a cold or flu drives resistance and doesn't help the patient.

Free-Response Practice

1. List the 4 types of pathogens and one example of each.

   Show answer

   Bacteria (E. coli), viruses (HIV), fungi (Candida), parasites (tapeworm).

2. What color does a gram-positive bacterium stain, and why?

   Show answer

   Purple — thick cell wall holds the crystal violet dye.

3. Why are gram-negative bacteria often harder to treat?

   Show answer

   Their outer membrane is a barrier that keeps many antibiotics from reaching their target.

4. List 3 factors that contribute to antibiotic resistance.

   Show answer

   Inappropriate use (for viruses), not finishing the course, agricultural use, broad-spectrum overuse, patient non-adherence.

5. What is the difference between culture and sensitivity?

Show answer

Culture identifies the bacteria. Sensitivity shows which antibiotic kills it.

Multiple Choice Practice

1. A patient has a fungal infection in the mouth. This is likely:
   A) Tinea · B) Candida (thrush) · C) Helminth · D) Virus

   Show answer

   B. Candida = oral thrush.

2. Viruses differ from bacteria because viruses:
   A) Have a thick cell wall · B) Stain with gram stain · C) Can't replicate without a host cell · D) Are killed by antibiotics

   Show answer

   C. Viruses need host cells to reproduce.

3. Which is a factor in antibiotic resistance? (SATA)
   A) Stopping antibiotics early · B) Using antibiotics for viral infections · C) Patient adherence · D) Agricultural antibiotic use · E) Hand hygiene

   Show answer

   A, B, D. Adherence and hand hygiene help PREVENT resistance.

Clinical Scenario

A patient finishes a 10-day course of broad-spectrum antibiotics for pneumonia and develops oral thrush and watery diarrhea. Explain what likely happened.

Show answer

The antibiotic wiped out normal flora, allowing opportunistic organisms to overgrow: **Candida** in the mouth (thrush) and possibly **C. difficile** in the gut (diarrhea). This is a classic complication of broad-spectrum antibiotic use.

☑ Self-Assessment Checklist

- I can name the 4 types of pathogens and an example of each.
- I can interpret gram stain (positive = purple, negative = pink, harder to treat).
- I can list 3 factors contributing to antibiotic resistance.
- I can distinguish superficial vs deep fungal infections.

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TOPIC 8

Stages of Infection, Transmission, & TB

MUST KNOW ▼

Routes of Transmission

- **Direct contact** — person to person (touch, sex). Examples: HIV, herpes, scabies.
- **Indirect contact** — fomites (contaminated objects). Example: doorknobs, IV poles.
- **Droplet** — large respiratory droplets (within ~3–6 feet). Examples: influenza, pertussis, COVID.
- **Airborne** — small particles travel long distances. Examples: TB, measles, varicella.
- **Vector-borne** — insects/animals. Examples: malaria (mosquito), Lyme (tick).
- **Vehicle/foodborne** — contaminated food or water.
- **Vertical (mother to baby)** — placenta, breast milk, birth canal.

Stages of Infection

Stage	What's happening	S/Sx
1. Exposure	Pathogen enters body	None yet
2. Incubation	Pathogen multiplies; person feels fine	Usually asymptomatic, but may be contagious
3. Prodrome	Early non-specific symptoms	Fatigue, low-grade fever, malaise; HIGHLY contagious
4. Acute (full)	Peak symptoms, immune response at max	Specific s/sx of the infection (rash, cough, etc.)
5. Convalescent	Body clears the pathogen, healing	Symptoms gradually resolve
6. Resolution	Recovery complete or chronic state established	No symptoms, or symptoms persist long-term

Tuberculosis (TB)

Bacterial infection by *Mycobacterium tuberculosis*; usually lives in the **lungs**. **Airborne** transmission.

Latent TB	Active TB
Immune system has walled off the bacteria. **No symptoms**. Not contagious. Positive skin test (PPD/ TST) but normal chest X-ray. Person is well.	Immune system weakened — TB escapes the wall. **Contagious**. Hemoptysis (coughing up blood), chest pain, productive cough > 3 weeks, fever, chills, weakness, fatigue, night sweats, weight loss.

Stages mnemonic: "*Every Infection Provides Acute Care Recovery*" — Exposure, Incubation, Prodrome, Acute, Convalescent, Resolution.

⚠ **Common confusion:** **Latent TB** = positive test but NO symptoms, NOT contagious. **Active TB** = symptoms AND contagious. Both need treatment, but the regimens differ.

Free-Response Practice

1. List the 6 stages of infection in order.

   Show answer

   Exposure → Incubation → Prodrome → Acute → Convalescent → Resolution.

2. During which stage is a person typically MOST contagious?

   Show answer

   Prodrome — early non-specific symptoms but pathogen is shedding heavily. Many transmit unknowingly here.

3. List 4 routes of transmission with an example of each.

   Show answer

   Direct contact (HIV); droplet (flu); airborne (TB, measles); vector-borne (malaria via mosquito); vehicle/foodborne (Salmonella); vertical (HIV mother → baby).

4. Compare latent vs active TB.

Show answer

Latent: walled off by immune system, no symptoms, NOT contagious, positive skin test. Active: bacteria escaped, symptoms (cough >3 wks, hemoptysis, fever, weight loss), contagious.

5. What route is TB transmitted by?

   Show answer

   Airborne (small respiratory particles).

Multiple Choice Practice

1. Which is an airborne pathogen?
   A) HIV · B) Tuberculosis · C) Salmonella · D) Lyme

   Show answer

   B. TB is airborne.

2. A patient with latent TB:
   A) Has active symptoms · B) Is highly contagious · C) Has a positive skin test but no symptoms · D) Should not be treated

   Show answer

   C. Positive PPD/TST, no symptoms, not contagious. Still treated to prevent activation.

3. Vague fatigue and low-grade fever, before specific symptoms appear, occur during which stage?
   A) Incubation · B) Prodrome · C) Acute · D) Resolution

   Show answer

   B. Prodrome — early non-specific symptoms.

Clinical Scenario

A patient with HIV develops a productive cough for 4 weeks, night sweats, and hemoptysis. PPD test is positive and chest X-ray shows infiltrates. Why is this patient at higher risk for active TB?

Show answer

HIV weakens the immune system (particularly CD4 T-helper cells), so the body can't keep latent TB walled off. The bacteria escape and replicate → active TB. Immunocompromised patients should be screened regularly and treated promptly for both latent and active TB.

☑ Self-Assessment Checklist

- I can name and order the 6 stages of infection.
- I can name 5 routes of transmission with examples.
- I can distinguish latent and active TB and identify TB's transmission route.
- I can list classic TB symptoms.

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TOPIC 9

UTIs: Cystitis vs. Pyelonephritis

MUST KNOW ▼

	Acute Cystitis (bladder infection)	Acute Pyelonephritis (kidney infection)
Location	Bladder (lower urinary tract)	Kidney (upper urinary tract)
Local s/sx	Dysuria, frequency, urgency, cloudy or bloody urine	Same plus **flank pain**
Systemic s/sx	**NO systemic symptoms** (typically no fever)	**YES — fever, malaise, nausea/vomiting**
Onset	Gradual	Abrupt
Complication	Can ascend to kidneys if untreated	**Urosepsis** possible (bacteria in bloodstream)
Lab findings	Positive urinalysis (WBCs, nitrites, leukocyte esterase); positive culture	Same plus possible elevated WBC, positive blood cultures, casts in urine

Special Presentation in Elderly

Older adults often present **differently** from textbook descriptions:

- Confusion, delirium, or change in mental status may be the ONLY sign of UTI.
- May not have classic dysuria or frequency.
- Falls, decreased function, urinary incontinence (new).
- Always rule out UTI in any elderly patient with new confusion.

Risk Factors

- Diabetes (glucose in urine feeds bacteria; impaired immune function).
- HIV/immunodeficiency.
- Anatomical abnormalities (kidney stones, vesicoureteral reflux).
- Female anatomy (shorter urethra closer to anus).
- Indwelling catheters (CAUTI).
- Pregnancy.
- Sexual activity.

Memory aid: Cystitis stays BELOW the belt = bladder symptoms only. Pyelonephritis goes ABOVE the belt = systemic symptoms (fever, flank pain).

⚠ **Common confusion:** A confused elderly patient with no obvious "UTI" symptoms may actually have one. Always check a urinalysis for new-onset confusion in older adults.

Free-Response Practice

1. List 3 symptoms of acute cystitis.

   Show answer

   Dysuria, frequency, urgency, cloudy/bloody urine; typically NO fever.

2. Which UTI typically presents with fever and flank pain?

   Show answer

   Pyelonephritis — kidney infection.

3. How does an elderly patient with a UTI typically present?

   Show answer

   Confusion or change in mental status may be the only sign — classic symptoms (dysuria, frequency) may be absent.

4. List 3 risk factors for UTI.

   Show answer

   Diabetes, immunodeficiency (HIV), anatomical abnormalities, female anatomy, catheters, pregnancy, sexual activity.

5. What is urosepsis?

   Show answer

   A bloodstream infection that originated from the urinary tract — complication of untreated pyelonephritis.

Multiple Choice Practice

1. The classic differentiator between cystitis and pyelonephritis is:
   A) Dysuria · B) Frequency · C) Fever and flank pain · D) Urine clarity

Show answer

C. Pyelo adds systemic symptoms (fever, flank pain, malaise, N/V).

2. Which is an "atypical" UTI presentation in the elderly? (SATA)
   A) New confusion · B) Falls · C) Classic dysuria · D) Urinary incontinence · E) Severe flank pain

   Show answer

   A, B, D. Confusion, falls, new incontinence are classic atypical presentations.

3. Untreated pyelonephritis can lead to:
   A) Otitis media · B) Urosepsis · C) Hepatitis · D) Hypothyroidism

   Show answer

   B. Urosepsis — bacteria spread from kidney to bloodstream.

Clinical Scenario

An 82-year-old woman is brought to the ER for "acting strange" and a fall. Vital signs: T 100.8°F, HR 110, BP 96/58. Labs show elevated WBC. She denies pain. Identify the most likely diagnosis and the priority assessment.

Show answer

Most likely: **UTI with possible urosepsis**. Elderly often present with confusion and falls rather than dysuria. The fever + tachycardia + hypotension suggest sepsis. Priority: **urinalysis, urine culture, blood cultures, IV fluids, and broad-spectrum antibiotics** while waiting for cultures.

☑ Self-Assessment Checklist

- I can list the local symptoms of cystitis.
- I can describe the systemic symptoms that distinguish pyelonephritis.
- I can identify atypical UTI presentations in the elderly.
- I can list 3 risk factors for UTI.
- I can define urosepsis.

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TOPIC 10

Immunity, WBCs, & Immunoglobulins (GAMED)

MUST KNOW

Innate vs Adaptive Immunity (the 3 Lines of Defense)

	INNATE (1st & 2nd line)	ADAPTIVE (3rd line)
Speed	Quick to respond	Slower to respond
Specificity	Non-specific	Specific to each antigen
Memory	No memory	**Has memory** (faster on re-exposure)
Components	Barriers (skin, mucous), secretions (saliva, tears, stomach acid), **macrophages** & **neutrophils**, inflammation, fever	**Lymphocytes** (B and T cells)
Key job	Block and engulf invaders quickly	Targeted attack & antibody production

Lymphocytes (Adaptive)

- **B cells** → produce **antibodies**. Antibodies = immunoglobulins = "GAMED."
- **T cells:**
  - **T helper cells (CD4)** — coordinate B and T cell responses; the "conductor."
  - **Cytotoxic T cells (CD8)** — directly destroy infected/foreign cells.

White Blood Cells (Leukocytes)

WBC	Function
Neutrophils	Most abundant; FIRST to arrive at bacterial infection; phagocytose
Lymphocytes	B & T cells; viral infection & adaptive immunity
Monocytes/ Macrophages	"Big eaters"; chronic infections, antigen presentation
Eosinophils	Parasites & allergies

Basophils	Release histamine; allergies

The 5 Immunoglobulins (Antibodies): GAMED

Ig	Key feature	Memory hook
IgG	Long-term immunity; crosses placenta mom → baby; most abundant	"**LonG-term, Gestational**"
IgA	In mucous membrane secretions (saliva, tears, breast milk); prevents entry	"**Access · Alimentary canal**"
IgM	First on the scene; in body fluids; biggest antibody	"**Monitor · First Mover**"
IgE	On mast cells; allergies, inflammation, parasites	"**AllergE**"
IgD	Found on naive B cells; function less clear (B-cell activation)	"**Don't know much about D**"

□**"GAMED" memory aids:** **G** = Gestational/lonG-term · **A** = Alimentary/Access (mucous) · **M** = Monitor/first · **E** = AllergE · **D** = Don't know much.

⚠ **Common confusion:** Innate immunity is FAST and NON-SPECIFIC. Adaptive immunity is SLOW and SPECIFIC with MEMORY. Don't reverse these.

□Free-Response Practice

1. Compare innate and adaptive immunity.

   Show answer

   Innate: fast, non-specific, no memory; barriers, macrophages, neutrophils. Adaptive: slower, specific, has memory; B and T cells.

2. Describe IgG, IgA, IgM, IgE, IgD in one short phrase each.

   Show answer

   IgG: long-term, crosses placenta. IgA: mucous secretions. IgM: first responder. IgE: allergies/parasites. IgD: naive B cells.

3. What do CD4 T helper cells do?

Show answer

Coordinate the immune response between B and T cells.

4. Which WBC is first to arrive at a bacterial infection?

Show answer

Neutrophil.

5. Which WBC and which Ig are involved in allergies?

Show answer

Eosinophils & basophils (WBCs) and IgE (antibody) bound to mast cells.

Multiple Choice Practice

1. Which antibody crosses the placenta to give the newborn passive immunity?
   A) IgA · B) IgM · C) IgE · D) IgG

Show answer

D. IgG — "Gestational."

2. The first antibody produced in a new infection is:
   A) IgG · B) IgA · C) IgM · D) IgE

Show answer

C. IgM — "Monitor/first responder."

3. Innate immunity includes which of these? (SATA)
   A) Skin · B) IgG antibodies · C) Neutrophils · D) T helper cells · E) Stomach acid

Show answer

A, C, E. IgG and T helpers are adaptive.

Clinical Scenario

A breastfeeding mother gets the flu shot. How does this protect both her and her newborn? Identify the antibody and the type of immunity transferred.

Show answer

The vaccine activates her adaptive immunity to make antibodies (mostly **IgG**) against the flu. **IgG** already crossed the placenta during pregnancy (passive). **IgA** in breast milk also protects the baby's mucous surfaces (passive). The baby's own immune system isn't generating these antibodies — it's borrowing mom's = **natural passive immunity**.

☑ Self-Assessment Checklist

- I can compare innate and adaptive immunity (speed, specificity, memory).
- I can name the major WBCs and the role of each.
- I can identify each of the 5 immunoglobulins (GAMED) with their key feature.
- I can describe what CD4 T helper cells do.

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TOPIC 11

⚠ Hypersensitivity, Anaphylaxis, Transplant & Immunity Types

MUST KNOW ▼

The 4 Types of Hypersensitivity Reactions

An over-reaction of the immune system to an antigen — the body's response damages tissue more than the antigen itself.

Type	Key Feature	Examples
Type I — "Allergic"	IgE on mast cells → histamine release. Immediate (mins).	Anaphylaxis, hay fever, asthma, food allergies, latex allergy
Type II — "antiBody"	Antibodies (IgG, IgM) target cells	Transfusion reaction (ABO mismatch), hemolytic disease of newborn, Goodpasture's
Type III — "Immune Complex"	Antigen-antibody complexes deposit in tissues	SLE (lupus), rheumatoid arthritis, glomerulonephritis
Type IV — "Delayed"	T-cell mediated. Hours to days.	TB skin test (PPD), poison ivy, contact dermatitis, transplant rejection

Anaphylaxis — A Type I Emergency

Massive IgE-mediated histamine release leading to:

- **Skin:** hives, flushing, swelling (angioedema).
- **Respiratory:** wheezing, stridor, throat tightening, airway compromise.
- **Cardiovascular:** hypotension (distributive shock), tachycardia.
- **GI:** nausea, vomiting, diarrhea.

Treatment: IM **epinephrine** (first-line), then oxygen, IV fluids, antihistamines, steroids, airway management.

Transplant Rejection

Type	Timing	Cause
Hyperacute		Pre-existing antibodies (Type II); ABO mismatch

	Minutes to hours	
Acute	Days to months	T-cell mediated (Type IV); most common; treated with immunosuppressants
Chronic	Months to years	Mixed mechanism; gradual organ damage

Graft-versus-host disease (GVHD): the OPPOSITE — donor T cells attack the recipient. Mostly seen with bone marrow / stem cell transplants.

Types of Immunity — How You Get Each

Type	How acquired	Example
Natural Active	Body makes antibodies after getting sick	Chickenpox infection → immune for life
Artificial Active	Body makes antibodies after vaccine	Flu shot, MMR, COVID vaccine
Natural Passive	Antibodies passed mom → baby	IgG across placenta, IgA in breast milk
Artificial Passive	Receiving someone else's antibodies	IVIG, post-exposure rabies/Hep B, monoclonal antibody infusions

Immunity mnemonic: "**A**ctive = **A**ctively making antibodies." "**P**assive = **P**re-made antibodies handed to you." Active = lasts a long time. Passive = short-term protection.

⚠ **Common confusion:** A TB skin test (PPD) is a **Type IV (delayed)** hypersensitivity, NOT Type I. The reaction develops over 48–72 hours.

Free-Response Practice

1. List the 4 types of hypersensitivity with one example each.

Show answer

I = Allergic (anaphylaxis); II = antiBody (ABO transfusion reaction); III = Immune complex (SLE); IV = Delayed (PPD, poison ivy, transplant rejection).

2. What are the major s/sx of anaphylaxis?

Show answer

Hives/angioedema, wheezing/stridor/airway tightening, hypotension/tachycardia, N/V/D. First-line: IM epinephrine.

3. What causes hyperacute transplant rejection?

Show answer

Pre-existing antibodies (Type II), often ABO mismatch — happens within minutes to hours of transplant.

4. Compare the 4 ways to get immunity.

Show answer

Natural active = catch the disease. Artificial active = vaccination. Natural passive = mom's antibodies (placenta, breast milk). Artificial passive = receiving antibody-containing products (IVIG, monoclonals).

5. Why does passive immunity last only a short time?

Show answer

The body isn't producing the antibodies. The borrowed antibodies degrade over weeks to months without immune memory being built.

□Multiple Choice Practice

1. A bee sting causing throat swelling and hypotension is which type of reaction?
   A) Type I · B) Type II · C) Type III · D) Type IV

   Show answer

   A. Type I (IgE-mediated, immediate) — anaphylaxis.

2. A patient who receives mismatched blood develops fever, back pain, and bloody urine. This is:
   A) Type I · B) Type II · C) Type III · D) Type IV

   Show answer

   B. Type II — antibody-mediated hemolytic transfusion reaction.

3. Receiving a tetanus vaccine produces which type of immunity?

A) Natural active · B) Artificial active · C) Natural passive · D) Artificial passive

Show answer

B. Artificial active — your body makes antibodies after vaccination.

Clinical Scenario

A patient receives a kidney transplant. Within 6 hours, the new kidney is failing. ABO testing confirms mismatch. What type of hypersensitivity is this, what's the rejection timing called, and what immune mechanism is responsible?

Show answer

Type II hypersensitivity (antibody-mediated). The rejection is **hyperacute**. Pre-existing antibodies in the recipient (anti-A or anti-B) bind donor kidney cells → complement activation → rapid destruction. This is why ABO and crossmatching are critical PRE-transplant.

☑ Self-Assessment Checklist

- I can name the 4 types of hypersensitivity with examples.
- I can describe anaphylaxis s/sx and first-line treatment.
- I can compare hyperacute, acute, and chronic transplant rejection.
- I can match the 4 types of immunity to how each is acquired.

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TOPIC 12

□HIV/AIDS, Immunodeficiency, Autoimmunity & Blood Cancers

MUST KNOW ▼

HIV & AIDS

HIV (Human Immunodeficiency Virus) is a **retrovirus** that targets **CD4 T helper cells** — the "conductor" of the immune response.

- As CD4 count drops, the immune system loses coordination.
- **AIDS** is the advanced stage: **CD4 count < 200** OR an AIDS-defining opportunistic infection.

Opportunistic Infections in AIDS

System	Examples
Respiratory	TB, PCP (Pneumocystis pneumonia)
GI	Oral thrush (Candida), esophageal candidiasis
CNS	Toxoplasmosis, cryptococcal meningitis
Cancers	Kaposi sarcoma, non-Hodgkin lymphoma

Immunodeficiency — Causes

- Cancer and cancer treatment (chemo, radiation).
- HIV infection.
- Diabetes (impaired neutrophil function, high glucose feeds organisms).
- Stress (chronic cortisol suppresses immunity).
- Malnutrition.
- Splenectomy (lose key immune organ).
- Immunosuppressive drugs (transplant, autoimmune disease).
- Age extremes (very young or very old).

Autoimmune Disease vs Immunodeficiency

	Autoimmune	Immunodeficiency
Problem	Immune system attacks own body	Immune system too weak to fight infections

Risk factors	Family history; women > men	HIV, chemo, diabetes, splenectomy, malnutrition
Examples	Lupus (SLE), RA, MS, Type 1 diabetes, Graves'	AIDS, SCID, post-transplant patients on immunosuppressants

Leukemia (Cancer of WBCs)

Leukemia = cancer of early blood cells. Abnormal proliferation; cells don't function.

- **Lymphoblastic** leukemias → affect cells that become **lymphocytes** (B and T).
- **Myelocytic** leukemias → affect cells that become **basophils, eosinophils, neutrophils**.
- **Acute** = rapid onset, immature cells, more aggressive.
- **Chronic** = slower, more mature cells, longer course.

Common features across leukemias: **fatigue, anemia, infections, bleeding/bruising, weight loss, splenomegaly**.

Lymphoma — Hodgkin vs Non-Hodgkin

	Hodgkin Lymphoma	Non-Hodgkin Lymphoma
Distribution	Localized; contiguous	Diffuse; non-contiguous
Defining cell	**Reed-Sternberg cells**	No Reed-Sternberg
Extranodal involvement	Uncommon	Common
Age	Younger adults / adolescents	Older adults
Prognosis	**Better — high survival rate** (early detection, localized, very treatable)	Worse

Hodgkin = "Has Reed-Sternberg, High survival." Non-Hodgkin = nO Reed-Sternberg, more diffuse, older patients, worse prognosis.

⚠ **Common confusion:** AIDS is defined by CD4 count < 200 OR an AIDS-defining opportunistic infection — not just having HIV. A person can have HIV for years without progressing to AIDS, especially with antiretroviral therapy.

Free-Response Practice

1. What cell does HIV target, and why is this devastating to the immune system?

   Show answer

   CD4 T helper cells. CD4 cells coordinate the entire adaptive immune response. Without them, B and other T cells can't function properly.

2. Define AIDS.

   Show answer

   Advanced HIV with CD4 < 200 cells/mm^3 OR the presence of an AIDS-defining opportunistic infection.

3. List 4 causes of immunodeficiency.

   Show answer

   Cancer/chemo, HIV, diabetes, malnutrition, stress, splenectomy, immunosuppressive drugs.

4. What's the key cell that distinguishes Hodgkin from Non-Hodgkin lymphoma?

   Show answer

   Reed-Sternberg cells (present in Hodgkin).

5. Why does Hodgkin lymphoma have a better prognosis?

   Show answer

   Localized, contiguous spread (detected earlier), affects younger patients (better treatment tolerance), highly responsive to chemotherapy and radiation.

Multiple Choice Practice

1. HIV is a:
   A) DNA virus · B) Retrovirus · C) Bacteria · D) Fungus

   Show answer

   B. Retrovirus — uses reverse transcriptase to incorporate into host DNA.

2. Reed-Sternberg cells are characteristic of:
   A) Acute leukemia · B) Hodgkin lymphoma · C) Non-Hodgkin lymphoma · D) Kaposi sarcoma

   Show answer

   B. Hodgkin lymphoma.

3. Which is an AIDS-defining illness? (SATA)
   A) Common cold · B) Kaposi sarcoma · C) PCP pneumonia · D) Oral thrush · E) Strep throat

   Show answer

   B, C. Kaposi sarcoma and PCP are AIDS-defining. Oral thrush is common in immunosuppression but not AIDS-defining alone.

Clinical Scenario

A 35-year-old with untreated HIV presents with shortness of breath, a CD4 count of 95, and chest X-ray showing diffuse infiltrates. Sputum confirms Pneumocystis jirovecii. Why has this patient developed PCP, and what is their diagnosis now?

Show answer

CD4 count below 200 makes the patient unable to fight off opportunistic infections like Pneumocystis (PCP). The combination of **CD4 < 200** plus an **AIDS-defining opportunistic infection** (PCP) means the patient has progressed from HIV to **AIDS**. Treatment: start antiretroviral therapy + treat PCP (TMP-SMX) + prophylaxis going forward.

☑ Self-Assessment Checklist

- I can describe what HIV targets (CD4) and define AIDS.
- I can name common opportunistic infections in AIDS.
- I can list 4+ causes of immunodeficiency.
- I can compare autoimmunity and immunodeficiency.
- I can compare lymphoblastic vs myelocytic leukemias and Hodgkin vs Non-Hodgkin lymphomas.

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Exam-Day Cheat Sheet

ABG Normals

pH 7.35–7.45 · CO_2 35–45 · HCO_3^- 22–26 · PaO_2 80–100

Electrolyte Normals

Na 135–145 · K 3.5–5.0 · Ca 9.0–10.5 · P 2.5–4.5 · Mg 1.3–2.1

ROME Trick

Respiratory Opposite, Metabolic Equal

Buffer Speeds

Serum (sec) → Respiratory (min) → Renal (days, STRONGEST)

RAAS Trigger

Low BP/volume/sodium → renin → angiotensin II → vasoconstriction + aldosterone + ADH + thirst

ANP/BNP

Opposite of RAAS — excrete Na & water. BNP marker of heart failure.

K^+ & pH

Acidosis → hyperkalemia. Alkalosis → hypokalemia.

EKG Changes

Hyperkalemia = peaked T waves. Hypokalemia = flat T + U waves.

Edema Causes

↑ hydrostatic, ↓ oncotic (low albumin), lymphatic block, capillary leak

Stages of Infection

Exposure → Incubation → Prodrome → Acute → Convalescent → Resolution

Gram Stain

Positive = PURPLE (thick wall). Negative = PINK (thin wall; harder to treat).

TB

Airborne. Latent = positive PPD, no symptoms. Active = contagious, cough >3 wk, hemoptysis.

UTIs

Cystitis = bladder, no fever. Pyelo = kidney, fever + flank pain. Elderly = confusion.

Immunity (Innate vs Adaptive)

Innate = fast, no memory. Adaptive = slow, has memory (B and T cells).

GAMED (Ig)

G=lonG/placenta. A=mucous secretions. M=first responder. E=allerGE/mast cells. D=naive B cells.

Hypersensitivity

I=Allergic/IgE. II=antiBody (transfusion). III=Immune complex (SLE). IV=Delayed (PPD, poison ivy).

HIV/AIDS

HIV targets CD4 T helpers. AIDS = CD4 <200 OR opportunistic infection.

Lymphoma

Hodgkin = Reed-Sternberg, localized, younger, BETTER prognosis. Non-Hodgkin = diffuse, older, worse.

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□Quick Glossary

Fluids & Hormones

- **ICF / ECF** — intracellular (high K^+) / extracellular (high Na^+).
- **Hydrostatic pressure** — pushing force from blood.
- **Oncotic pressure** — pulling force from albumin.
- **Edema** — fluid in tissues from pressure imbalance, lymphatic block, or capillary leak.
- **RAAS** — renin → angiotensin II → aldosterone; raises BP.
- **Aldosterone** — reabsorbs Na & water, excretes K.
- **ADH** — reabsorbs water only.
- **ANP / BNP** — opposite of RAAS; excrete Na & water.

Electrolytes

- **Hyponatremia / Hypernatremia** — brain affects: confusion, seizures.
- **Hypokalemia / Hyperkalemia** — cardiac dysrhythmias; EKG changes.
- **Hypocalcemia** — Trousseau's, Chvostek's, tetany.
- **Hypercalcemia** — "bones, stones, groans, psychic moans."
- **Trousseau's sign** — carpal spasm with BP cuff.
- **Chvostek's sign** — facial twitch when cheek tapped.

Acid-Base & ABGs

- **Acidosis** — pH < 7.35. **Alkalosis** — pH > 7.45.
- **CO_2 = acid**. **HCO_3^- = base**.
- **Kussmaul respirations** — deep rapid breathing in DKA.
- **Buffer systems** — serum, respiratory, renal.
- **Anion gap** — high in DKA, lactic acidosis, salicylates.
- **Compensation** — the OTHER value shifts to bring pH back toward normal.

Pathogens & Infection

- **Gram positive** — thick wall, purple. **Gram negative** — thin wall, pink, harder to treat.
- **Antibiotic resistance** — misuse, incomplete courses, agricultural use.
- **TB latent vs active** — walled off vs spreading; only active is contagious.
- **Stages of infection** — Exposure → Incubation → Prodrome → Acute → Convalescent → Resolution.
- **Cystitis** — bladder infection, no fever. **Pyelonephritis** — kidney, fever + flank pain.
- **Urosepsis** — bloodstream infection from UTI.

Immune System

- **Innate** — fast, non-specific, no memory. **Adaptive** — slow, specific, memory.
- **B cells** — make antibodies. **T cells** — CD4 helpers, CD8 cytotoxic.
- **WBCs** — neutrophils, lymphocytes, monocytes, eosinophils, basophils.
- **Immunoglobulins (GAMED)** — G long-term, A access/mucous, M first responder, E allergy, D naive B cells.
- **Hypersensitivity I–IV** — allergic, antiBody, immune complex, delayed.
- **Anaphylaxis** — Type I; treat with IM epinephrine.
- **Immunity types** — natural/artificial × active/passive.

Immune Disorders & Cancers

- **HIV** — retrovirus targeting CD4 cells.
- **AIDS** — CD4 < 200 or AIDS-defining opportunistic infection.
- **Opportunistic infections** — PCP, oral thrush, toxoplasmosis, Kaposi sarcoma.
- **Autoimmune** — immune attacks self (lupus, RA, Type 1 DM).
- **Immunodeficiency** — immune too weak (HIV, chemo, malnutrition, splenectomy).
- **Leukemia** — cancer of WBC precursors; lymphoblastic (B/T) or myelocytic.
- **Hodgkin lymphoma** — Reed-Sternberg cells, localized, better prognosis.
- **Non-Hodgkin lymphoma** — diffuse, no Reed-Sternberg, older, worse prognosis.

Final Study Tips for BIO 280 Exam 1

- **Memorize lab values first.** If you don't know the normal ranges, you can't recognize abnormal patterns.
- **Practice ABGs until they're automatic.** Do all 4 worked examples, then make up your own.
- **Make a single-page table** with electrolytes vs. high/low s/sx side-by-side. Re-write it without looking.
- **Connect mechanisms to s/sx.** Don't memorize lists — understand WHY hyperkalemia peaks T waves, WHY hypocalcemia causes tetany.
- **Practice with the hidden answers.** Active recall beats passive re-reading every time.
- **Use the cheat sheet** the night before AND morning of the exam.
- **Set up your testing environment** the day before — lock down browser installed, room scanned, whiteboard ready, no devices.
- **Don't rush.** 50 questions, plenty of time. Read each option carefully — SATA needs ALL correct answers.

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In your web browser, press **Ctrl+P** (Windows) or **Cmd+P** (Mac). Choose "Save as PDF." All collapsed topics will automatically expand so the printed version is complete.

You've got this. Modules 1 & 2 are dense, but the patterns are learnable. Trust the work.